

Jose Cuaran

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EDUCATION

- 2021 - Dec-2026* **PhD in Computer Science** at **University of Illinois Urbana-Champaign, USA**
* Expected **Advisor:** Girish Chowdhary (Distributed and Autonomous Systems Laboratory)
Dissertation: *Active Semantic Mapping and Task Planning for Autonomous Agricultural Robots*
Relevant Coursework: Computer Vision, 3D Vision, Mobile Robotics, Robot Learning, Deep Learning for Robotic Manipulation, Deep Learning, Machine Learning
GPA: 3.9/4.0
- 2014 - 2016 **M.S. in Electrical Engineering** at **Universidad Nacional De Colombia**
GPA: 4.3/5.0
- 2007 - 2013 **B.S. in Electrical Engineering** at **Universidad Nacional De Colombia**
GPA: 4.3/5.0 Class Rank: 1/33
- 2007 - 2012 **B.S. in Electronics Engineering** at **Universidad Nacional De Colombia**
GPA: 4.3/5.0 Class Rank: 2/47

RESEARCH EXPERIENCE

- Aug. 2021 to Present **Ph.D. Graduate Research Assistant**
University of Illinois at Urbana-Champaign, Urbana, IL, USA
- Conduct research at the intersection of active perception, semantic mapping, and language-guided autonomy for robotic systems operating in unstructured environments.
 - Investigated the integration of semantic maps with visual-language models to enable flexible, [language-guided task execution for autonomous crop monitoring](#) (Cuaran, Koe, et al. [2026](#)).
 - Integrated classical map representations with recent neural methods for [high-fidelity reconstruction of plants with mobile manipulators](#) (Cuaran, Upalapati, and Chowdhary [2026](#)).
 - Developed an [active semantic mapping framework for mobile manipulators](#) in horticultural environments that selects informative viewpoints under occlusion and uncertainty, significantly improving exploration efficiency (Cuaran, Ahluwalia, et al. [2025](#)).
 - Introduced one of the first [large-scale, multi-modal under-canopy datasets](#) for row crops, enabling systematic evaluation of state-of-the-art SLAM systems and revealing critical performance gaps (Cuaran, Baquero Velasquez, et al. [2024](#)).
 - Validated methods through real-world deployments on robotic platforms in agricultural environments.

PROFESSIONAL EXPERIENCE

- Feb. 2017 to Jul. 2017 **Electrical Design Engineer**
Disenos Electricos Julio Cesar Garcia, Bogota D.C.
Designed electrical systems for residential and commercial installations and substations, including electrical calculations, service entrance sizing, and technical drawings.

TEACHING EXPERIENCE

Aug. 2025 to Present	Graduate Teaching Assistant University of Illinois at Urbana-Champaign, Urbana, IL, USA Department of Computer Science - Supported CS543 – <i>Computer Vision</i> - Updated and graded assignments
Aug. 2022 to Dec. 2022	Graduate Teaching Assistant University of Illinois at Urbana-Champaign, Urbana, IL, USA Department of Computer Science - Supported CS598 – <i>Mobile Robotics</i> - Updated assignments and coding exercises from ROS1 to ROS2 - Implemented autograding for assignments.
Mar. 2014 to May. 2016	Graduate Teaching Assistant Universidad Nacional de Colombia, Bogota D.C. Department of Electrical and Electronics Engineering - Supported undergraduate courses on <i>Signal and Dynamic Systems</i> and <i>Automatic Control</i> . - Developed and graded quizzes, exams, and homeworks. - Designed laboratory guides.

OTHER ACTIVITIES AND COURSES

2021	Deep Learning Specialization (Coursera)
2020	Machine Learning (Coursera)
Sep 03 to Sept 20, 2019	Malaysian Technical Cooperation Programme Application and Development of PLC Systems Centre for instructor and advanced skill training (CIAST)(Selangor - Malaysia)
Jun 10 to Jun 13, 2019	Production processes in industry 4.0 Production and logistics management FESTO (Bogota D.C.)
Jan 21 to Jan 25, 2019	Mobile and Fixed Robotics FESTO (Bogota D.C.)

HONORS AND AWARDS

August, 2020	Fulbright Scholarship for Doctoral Studies Minciencias and Fulbright Commission of Colombia
Oct, 2020	Teaching Excellence Award Universidad Católica de Colombia
May, 2019	Sense of Being a Teacher Award Universidad Católica de Colombia
Jun 2016	Meritorious Distinction of Master Thesis Universidad Nacional de Colombia
March, 2014—May, 2016	Teaching Assistant Scholarship Universidad Nacional de Colombia
Feb, 2014	Direct Admission to a Graduate Program Universidad Nacional de Colombia

PROJECTS

Generative Depth Completion. Training diffusion models to generate smooth, metrically accurate depth maps from noisy RGB-D observations.

Learning to Turn. Training a diffusion policy for autonomous row-to-row navigation of a ground robot in corn fields.

TagSLAM. Developing a factor-graph-based localization system that fuses visual odometry with AprilTag observations for drift correction.

TECHNICAL SKILLS

Programming	Python, C++
AI/ML	PyTorch, Vision Foundation Models, VLMs, Diffusion Models
Computer Vision	3D Reconstruction, Semantic Mapping, Neural Scene Representations
Robotics	ROS/ROS2, SLAM, State Estimation, Sensor Fusion, Motion Planning
Libraries	NumPy, Open3D, PCL, GTSAM
Tools & Systems	CUDA, Linux, Docker, Git, Conda
Platforms	NVIDIA Jetson, Raspberry Pi, Arduino

LANGUAGES

Spanish	Native
English	Advanced

PUBLICATIONS

Cuaran, Jose, Kendall Koe, et al. (2026). “Visual-language-guided task planning for horticultural robots”. In: *Computers and Electronics in Agriculture* 252, p. 112058.

Cuaran, Jose et al. (2026). “Active Semantic Mapping of Horticultural Environments Using Gaussian Splatting”. In: *arXiv preprint arXiv:2601.12122*. Under review.

Cuaran, Jose, Kulbir Singh Ahluwalia, et al. (2025). “Active Semantic Mapping with Mobile Manipulator in Horticultural Environments”. In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, pp. 12716–12722.

Cuaran, Jose, Andres Eduardo Baquero Velasquez, et al. (2024). “Under-canopy dataset for advancing simultaneous localization and mapping in agricultural robotics”. In: *The International Journal of Robotics Research* 43.6, pp. 739–749.

Cuaran, Jose and Jose Leon (2021). “Crop monitoring using unmanned aerial vehicles: A review”. In: *Agricultural Reviews* 42.2, pp. 121–132.